

Two-Sample t-Test

Audity Ghosh

Understanding and Applying
Hypothesis Testing on Two Datasets

Objectives

- - Understand the concept of hypothesis testing using a two-sample t-test
- - Apply the test to compare means of two independent samples

Theory

- The two-sample t-test determines whether the means of two independent samples are significantly different.
- Types:
 - 1. Equal Variance (Pooled t-test)
 - 2. Unequal Variance (Welch's t-test)

Equal Variance t-test (Pooled t-test)

- The **pooled t-test** assumes that the two population have the **same variance** (homogeneity of variance). It pools the variances from both samples to estimate a common variance.

Equal Variance t-test (Pooled t-test)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where:

- $\bar{X}_1, \bar{X}_2$ are the sample means,
- s_p^2 is the **pooled variance**, calculated as:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

- n_1, n_2 are sample sizes,
- s_1^2, s_2^2 are the variances of each sample.

Equal Variance t-test (Pooled t-test)

- **Example: Testing Exam Scores of Two Classes**
- Suppose we compare the exam scores of **Class A** and **Class B**, assuming their performance variability is similar.
- **Class A scores:** [85, 90, 88, 92, 87, 89, 91, 86, 90, 88]
- **Class B scores:** [78, 82, 80, 79, 81, 83, 77, 82, 80, 79]
- We assume **equal variance** and apply the pooled t-test.

Unequal Variance t-test (Welch's t-test)

- The **Welch's t-test** is used when the two groups **do not have the same variance**. It is more flexible and does not assume equal variances.

Two Sample T-Test

- The **two-sample t-test** (also known as the **independent t-test**) is a statistical test used to *compare the means of two independent groups to determine whether they are significantly different from each other.*

Key Assumptions

- The two samples are independent.
- The data in each group are normally distributed
- The variances of the two groups are equal (for the standard t-test) or unequal (for the Welch's t-test).

Hypothesis Testing Framework

- Null Hypothesis (H_0): The means of the two samples are equal ($\mu_1 = \mu_2$)
- Alternative Hypothesis (H_1): The means are not equal ($\mu_1 \neq \mu_2$)
- Significance Level (α): Commonly set to 0.05

Degrees of Freedom (df) in the Two-Sample t-Test

- Degrees of freedom (df) represent the number of independent values that can vary in a statistical calculation. In the context of the **two-sample t-test**, **df helps determine the shape of the t-distribution** used to calculate the p-value.
- The calculation of degrees of freedom **differs** between the **Equal Variance (Pooled t-test)** and **Unequal Variance (Welch's t-test)**.

Degrees of Freedom (df) in the Two-Sample t-Test

For the Unequal Variance (Welch's t-test), the degrees of freedom are not fixed but are estimated using the Welch-Satterthwaite equation:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

where:

- s_1^2, s_2^2 are the sample variances,
- n_1, n_2 are the sample sizes.

Working Procedure

- 1. Define H_0 and H_1
- 2. Collect and define two independent datasets
- 3. Specify variance assumption (equal/unequal)
- 4. Perform the t-test using Python
- 5. Interpret results (t-statistic, p-value, confidence interval)

Example: Testing the Effectiveness of a New Drug

- **Scenario:**

A pharmaceutical company wants to test whether a new drug (**Drug A**) is more effective in reducing blood pressure compared to a Amdocol Plus 50 (**Drug B**).

- **Group 1 (Drug A group):** 10 patients took Drug A, and their blood pressure was measured.
- **Group 2 (Amdocol Plus 50 group):** 10 different patients took Amdocol plus 50, and their blood pressure was also measured.

Example: Testing the Effectiveness of a New Drug

- **Collected Data (Systolic Blood Pressure after Treatment):**
 - **Drug A group:** [120, 115, 118, 122, 119, 117, 116, 121, 120, 118]
 - **Amdocol group:** [130, 128, 132, 135, 129, 127, 126, 134, 133, 131]

Example: Testing the Effectiveness of a New Drug

- **Step 1: State the Hypotheses**
 - **Null Hypothesis (H_0):** There is no difference in the mean blood pressure reduction between the two groups. ($\mu_1 = \mu_2$)
 - **Alternative Hypothesis (H_1):** There is a significant difference in the mean blood pressure reduction between the two groups. ($\mu_1 \neq \mu_2$)

Example: Testing the Effectiveness of a New Drug

Step 2: Perform the Two-Sample t-test

The test statistic is calculated using:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Drug A and Amdocol groups.
- s_1^2, s_2^2 are the sample variances.
- n_1, n_2 are the sample sizes.

Example: Testing the Effectiveness of a New Drug

- **Step 3: Interpret the Results**
- If the **p-value** from the test is **less than** the significance level (usually 0.05), we **reject the null hypothesis**, meaning there is a significant difference in mean blood pressure reduction between Drug A and the placebo.
- If the **p-value** is **greater than** 0.05, we **fail to reject the null hypothesis**, meaning there is no significant difference between the two groups.

Results

- **Results of the Two-Sample t-Test**

- t-statistic= -10.02
- p-value = 8.64×10^{-9} (very small)

- **Interpretation**

- Since the p-value is much smaller than the common significance level (0.05), we reject the null hypothesis (H_0). This means there is a statistically significant difference between the mean blood pressure levels of the Drug A group and the Amdocol group.
- In simpler terms, **Drug A has a significant effect in lowering blood pressure compared to the Amdocol**.

Interpretation of Results

- - $p\text{-Value} < 0.05 \rightarrow$ Reject H_0 (Significant difference in means)
- - $p\text{-Value} \geq 0.05 \rightarrow$ Fail to reject H_0 (No significant difference)
- - Confidence interval helps assess the mean difference

Conclusion of Scenario

- If the test results show a significant difference, the company can conclude that Drug A is effective in reducing blood pressure compared to the Amdocol.
- If not, there is no strong statistical evidence to support the claim that Drug A is better.

Two-Sample t-Test: Comparing Two Independent Means

- The **two-sample t-test** is a statistical method used to compare the means (averages) of **two independent groups** and determine whether the difference between them is **statistically significant**.
 - ♦ **Independent samples** mean that the groups do not affect each other.
 - ♦ This test helps answer the question: "**Is there a real difference between the two groups, or is it just due to random chance?**"

Two-Sample t-Test: Comparing Two Independent Means

- **Example:**
- Suppose we want to check if a new diet plan helps in weight loss. We take **two groups of people**:
- Group 1 follows the **new diet plan**.
- Group 2 follows their **regular diet**.
- We measure their **average weight loss** after one month and perform a **two-sample t-test** to check if the difference in average weight loss is significant.

p-Value: Understanding Statistical Significance

- The **p-value** tells us how likely we are to get our test results **if the null hypothesis (H_0) is true**.
- ♦ **Null Hypothesis (H_0):** There is **no real difference** between the means of the two groups.
 - ♦ **Alternative Hypothesis (H_1):** There is a **real difference** between the means.

p-Value: Understanding Statistical Significance

- **Key Interpretation of the p-value:**
 - If $p > 0.05$, we **fail to reject** $H_0 \rightarrow$ **No significant difference** between the groups.
 - If $p < 0.05$, we **reject** $H_0 \rightarrow$ **Significant difference** between the groups.
- **Example:**
- If the p-value is **0.02**, it means there is a **2% chance** that the observed difference in weight loss happened by random chance. Since **$0.02 < 0.05$** , we **reject the null hypothesis** and conclude that the diet **significantly affects weight loss**.

Confidence Interval (CI): Estimating the True Difference

- The **confidence interval (CI)** gives us a range of values where we expect the true difference in means to lie.
-  **If the CI includes 0** → The difference is **not statistically significant**.
-  **If the CI does not include 0** → The difference is **statistically significant**.

Confidence Interval (CI): Estimating the True Difference

- **Example:**
- If the **95% confidence interval** for weight loss difference is **(1.5 kg, 4.2 kg)**:
- Since **0 is not in this range**, we conclude that the diet **has a significant effect**.
- We are **95% confident** that the true difference in weight loss between the groups is between **1.5 kg and 4.2 kg**.
- However, if the CI was **(-0.5 kg, 2.3 kg)**:
- Since **0 is in this range**, we **cannot confidently say** the diet is effective.

Variances: Why We Used the Unequal Variance t-Test (Welch's t-Test)

- ♦ **Variance** measures how much the data points in each group **spread out** from the mean.
 - ♦ If the two groups have very **different variances**, the standard t-test may **not be reliable**.
 - ♦ **Welch's t-test** is more **robust** and does **not assume equal variances**.

Variances: Why We Used the Unequal Variance t-Test (Welch's t-Test)

- **Example:**
- **Group 1 (Diet plan):** Weight loss values vary a **lot** (some lose 5kg, some lose 0.5kg).
- **Group 2 (Regular diet):** Weight loss values are more **consistent**.
- Since the two groups have **different variances**, the **Welch's t-test** is the better choice.

Importance of Choosing the Right t-Test

- This experiment shows that choosing the **correct version of the t-test** is essential.
- ♦ If variances are **equal**, we use the **pooled t-test** (Student's t-test).
 - ♦ If variances are **unequal**, we use **Welch's t-test**.
 - ♦ Interpreting the **p-value and confidence interval** helps in decision-making.

Importance of Choosing the Right t-Test

- **Example Decision:**
- If our t-test **proves the diet plan is effective**, the company can confidently promote it.
If the t-test **shows no significant difference**, they may need to **rethink their marketing**.

Summary

Concept	Meaning	Example Interpretation
Two-Sample t-Test	Compares the means of two groups	Does a new diet plan cause more weight loss than a regular diet?
p-Value	Probability of getting results if no real difference exists	$p = 0.02$ means there is only a 2% chance the result is random.
Confidence Interval	Range of likely values for the true mean difference	(1.5kg, 4.2kg) means the diet likely helps with weight loss.
Variances	Measures how spread out the data is	Welch's t-test is used if one group's weight loss varies more than the other.
Choosing the Right t-Test	Helps make reliable decisions	Use pooled t-test for equal variances, Welch's for unequal.

Conclusion

- The two-sample t-test is a powerful method to compare means of two independent samples. Understanding variance assumptions and interpreting results correctly is crucial for data-driven decision-making.